

Since, **b** is the only character left after performing all the operations. Therefore, **b** is returned as the output.

Example 2:

input1 : j#k&h
input2 : 5

Output : &

Explanation:

The following operations can be performed on the string "j#k&h":

- $s = j \# k \& h j \# k \& h j \# k \& h j \# k \& h j \# k \& h$ (The string obtained after appending the given string K-1 times)
- $s = \# \& j k h \# \& j k h \# \&$ (The string obtained after removing every alternate character from the beginning)
- $s = \# j h \& k \#$ (The string obtained after removing every alternate character from the end)
- $s = j \& \#$ (The string obtained after removing every alternate character from the beginning)
- $s = \&$

& is the only character left after performing all the operations. Therefore, **&** is returned as the output.

Output specification:

Return a string representing the last remaining character after performing all the operations mentioned.

Example 1:

input1 : abcd

input2 : 3

Output : b

**Explanation:**

The following operations can be performed on the string "abcd":

- $S = a b c d a b c d a b c d$ (The string obtained after appending the given string $K-1$ times)
- $S = b d b d b d$ (The string obtained after removing every alternate character from the beginning)
- $S = b b b$ (The string obtained after removing every alternate character from the end)
- $S = b$

Since, b is the only character left after performing all the operations. Therefore, b is returned as the output.

Example 2:

Question 2

Revisit Later

How to Attempt?

Last One Standing

You are given an alphanumeric string **S** and an integer **K**. Consider a new string in which S is appended K-1 times.

Now, in this new string, the following operations are performed:-

1. Every alternate character starting from the first character is removed.
2. Every alternate character starting from the last character is removed.

The above two operations are repeated until one character remains. Your task is to find and return a string representing the last remaining character after performing all the operations.

Note: Here, alphanumeric refers to the string that may contain alphabet (a-z and A-Z), numerals (0-9), and certain special characters such as '\$', '#', '&', and '*'.

Input Specification:

input1 : A string S, representing the alphanumeric string.

input2 : An integer value K.

Output Specification:

Return a string representing the last remaining character after performing all the operations mentioned.



output array will be $[1, 0, 2, 7, 0]$ and here the total power produced by the solar panels in a day will be:

$$(1*1) + (2*6) + (3*3) + (4*4) + (5*8) = 78.$$

No other orientation of these solar panels can exceed this total power production. Hence, **78** will be returned as the output.

Example 2:

input1 : 4

input2 : {3,5,-9,10}

Output : 52

Explanation:

Originally, the energy output of the solar panels is {3,5,-9,10} and the total power that the solar panels can produce in a day is:

$$(1*3) + (2*5) + (3*-9) + (4*10) = 26.$$

To maximize the total power production of the solar panels, Bruce can reposition the second last solar panel {-9} before the first solar panel {3}. The new energy output array will be {-9,3,5,10} and here the total power produced by the solar panels in a day will be:

$$(1*-9) + (2*3) + (3*5) + (4*10) = 52.$$

No other orientation of these solar panels can exceed this total power production. Hence, **52** will be returned as the output.

CPP

1	#
2	#
3	/
4	
5	i
6	{
7	
8	
9	
10	
11	
12	
13	}

☐ Use Custom

Explanation:

Originally, the energy output of the solar panels is {8,1,6,3,4} and the total power that the solar panels can produce in a day is:

$$(1*8) + (2*1) + (3*6) + (4*3) + (5*4) = 60.$$

To maximize the total power production of the solar panels, Bruce can reposition the first solar panel {8} after the last solar panel {4}. The new energy output array will be {1,6,3,4,8} and here the total power produced by the solar panels in a day will be:

$$(1*1) + (2*6) + (3*3) + (4*4) + (5*8) = 78.$$

No other orientation of these solar panels can exceed this total power production. Hence, **78** will be returned as the output.

Example 2:

input1 : 4

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Explanation:

Originally, the energy output of the solar panels is {3,5,-9,10} and the total power that the solar panels can produce in a day is:

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To maximize the total power production of the solar panels, Bruce can

CPP

1

2

3

4

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6

7

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11

12

13



Input Specification:

input1 : An integer value N, representing the number of solar panels.
input2 : An integer array X, representing the energy output of N solar panels.

Output Specification:

Return an integer value representing the maximum total power that the solar panels can produce in a day.

Example 1:

input1 : 5
input2 : {8,1,6,3,4}

Output : 78

Explanation:

Originally, the energy output of the solar panels is {8,1,6,3,4} and the total power that the solar panels can produce in a day is:

$$(1*8) + (2*1) + (3*6) + (4*3) + (5*4) = 60.$$

To maximize the total power production of the solar panels, Bruce can reposition the first solar panel {8} after the last solar panel {4}. The new energy output array will be {1,6,3,4,8} and here the total power produced by the solar panels in a day will be:

$$(1*1) + (2*6) + (3*3) + (4*4) + (5*8) = 78$$



Question 1

 Revisit Later

CPP

How to Attempt?

Sun City

Bruce is the energy administrator in a city. He has set up **N** solar panels throughout the city which generate varying amounts of energy. The energy output of **N** solar panels is given in an integer array **X**.

The power output of a single solar panel in a day can be calculated by multiplying the energy output of a solar panel with its corresponding index. Bruce can reposition a solar panel by placing it either before or after another solar panel, but he can perform this repositioning only once as it is expensive to reinstall a solar panel.

Your task is to help Bruce find and return an integer value representing the maximum total power that the solar panels can produce in a day.

Note:

- Assume 1 based indexing.
- The energy output of a solar panel can be either negative or positive.

Input Specification:

input1 : An integer value **N**, representing the number of solar panels.

input2 : An integer array **X**, representing the energy output of **N** solar panels.

☐ Use C